

Towards Trauma-Informed Data Donation of Sexual Experience in Online Dating to Improve Sexual Risk Detection AI

Wenqi Zheng*
Emma Walquist
Psychology, Oakland
University
USA, Rochester

Isha Datey
Computer Science and
Engineering, Oakland
University
USA, Rochester

Xiangyu Zhou
Computer Science, Wayne
State University
USA, Detroit

Kelly Berishaj
Nursing, Oakland
University
USA, Rochester

Melissa McDonald
Michele Parkhill
Psychology, Oakland
University
USA, Rochester

Dongxiao Zhu
Computer Science, Wayne
State University
USA, Detroit

Douglas Zytke
College of Innovation &
Technology, University of
Michigan-Flint
USA, Flint

ABSTRACT

Sexual risk detection AI has been touted as a scalable solution for computer-mediated sexual violence. Data donation is a user-centered approach to producing ecologically valid datasets for sexual risk detection AI: voluntarily providing personal data that is representative of risk. However, the act of donating intimate sexual experience data can itself be traumatizing. We propose *Ube*: a trauma-informed sexual experience data donation app for online daters that is developed jointly with sexual violence experts and care practitioners. Cognitive walkthroughs of *Ube* with these experts elucidated several design approaches to mitigate retraumatization during data donation, including a conversational agent and mental health checks.

CCS CONCEPTS

• **Human-centered computing** → Walkthrough evaluations; • **Applied computing** → Psychology.

KEYWORDS

Sexual Violence, Data Donation, Cognitive Walkthrough, Online Dating, Harm, AI, Risk Detection

ACM Reference Format:

Wenqi Zheng, Emma Walquist, Isha Datey, Xiangyu Zhou, Kelly Berishaj, Melissa McDonald, Michele Parkhill, Dongxiao Zhu, and Douglas Zytke. 2023. Towards Trauma-Informed Data Donation of Sexual Experience in Online Dating to Improve Sexual Risk Detection AI. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3616689>

*Both authors contributed equally to this research.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3616689>

1 INTRODUCTION

Risk detection AI—the use of artificial intelligence (AI) to automatically detect and mitigate risk of various computer-mediated harms—has emerged as a scalable solution [5, 16, 22, 27] to computer-mediated sexual violence (SV). Data donation and annotation [4, 6, 12, 21] have emerged as popular approaches to provide ecologically valid datasets [3, 22] for model training of risk detection AI that represent real experiences of harm.

Data donation for sexual risk detection [20] may involve disclosure of intimate (and harmful) sexual experiences, so the very act of data donation can be traumatizing to users. We explore how to identify and mitigate adverse impacts incurred through the act of sexual experience data donation through in-progress development of *Ube*: a mobile app for donating data about sexual experiences that occur through online dating (a now-well-known context for online-to-offline SV [2, 11, 14, 28]). *Ube* is developed jointly with SV experts and care practitioners who have several years of experience directly interacting with victims and perpetrators of SV so as to translate to app design their trauma-informed practices [9, 23] to maintain comfort during the act of reporting sexual experiences.

In the following sections we introduce the *Ube* prototype and findings from six **cognitive walkthroughs** [26] to evaluate the app with SV experts in accordance with two research questions: **RQ1**. *What are the harms that may occur through the act of donating sexual experience data?* **RQ2**. *How can data donation platforms be designed to mitigate such harms?*

2 PROTOTYPE OVERVIEW & DESIGN

Ube is a mobile application that uses a conversational agent to guide users in donating data about their recent online-to-offline sexual experience with others, such as messaging interactions, face-to-face dates, and sexual activity. Donation takes various forms including: uploading behavioral data (e.g., screenshots from the dating app, JSON files of messaging interactions), and answering multiple-choice and open-ended questions to lend context to the donated data, and self-report of experiences that behavioral data is not available for (e.g., face-to-face encounters). A semi-interactive prototype for *Ube* was first developed in Figma; we then moved to

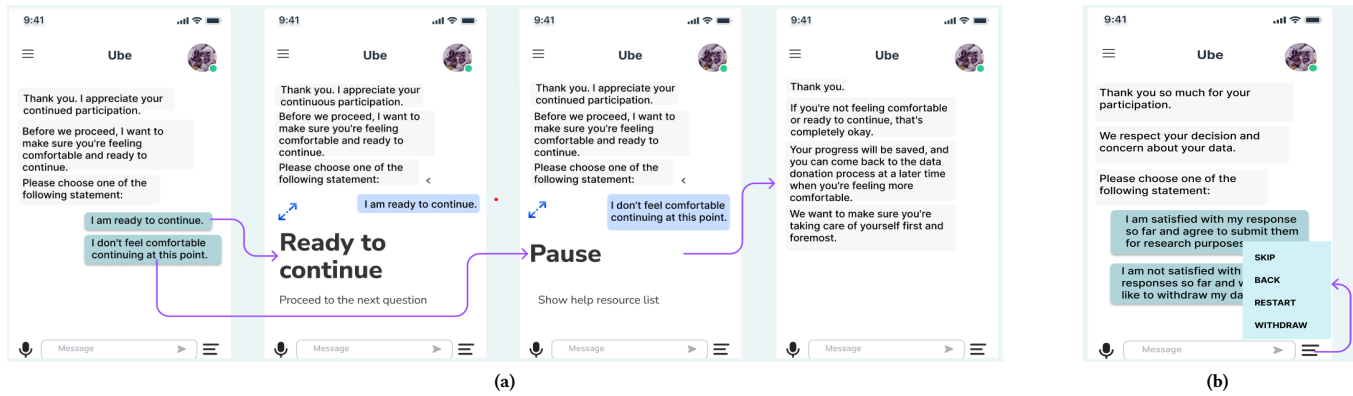


Figure 1: 1A shows the check-in which participants are shown before being asked questions that are most likely to be retraumatizing. 1b shows users can click the hamburger icon on the bottom right to skip, restart, or withdrawal from the study.

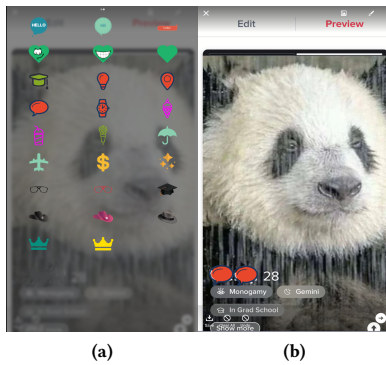


Figure 2: 2(a) shows users can manually cover the Tinder profile page they donate using the provided stickers; 2(b) shows part of the screenshot covered by two stickers.

develop *Ube* on NPM, a renowned package manager for JavaScript, for full interactivity and fidelity.

2.1 Evaluation with Sexual Violence Experts

Our design team consists of two UX design leads and six design consultants with SV expertise: one Sexual Assault Nurse Examiner (SANE), two SV shelter workers, and three SV researchers—two in psychology, and one in HCI. The design process entailed weekly meetings with SV experts to generate activity scenarios [24] articulating through text an exhaustive list of possible data donation journeys and a semi-interactive prototype for *Ube* in Figma. The prototype was then subjected to iterative evaluation through six rounds of cognitive walkthroughs using Spencer’s streamlined method [26]. Finally, each cognitive walkthrough was video-recorded, transcribed, and analyzed in Nvivo following Braun and Clarke’s thematic analysis approach [7].

3 FINDINGS

Analysis of the cognitive walkthroughs for our data donation application *Ube* revolved around an overarching theme of applying *trauma-informed approach* [15]: (R)ecognizing that trauma is common and can have a significant impact; (R)ecognizing signs and symptoms of trauma; and avoiding (R)etraumatization.

Minimizing Retraumatization During Data Donation. A notable cause of secondary harm is retraumatization; relieving stress reactions experienced from a previously harmful event when presented with a new but similar situation. This can include symptoms such as flashbacks and avoidance [8, 19, 23]. *Ube* is designed as a chatbot that scaffolds a comfortable and sympathetic environment for self-disclosure, avoiding insensitive questioning and implicit victim-blaming. We include randomly interspersed questions that prompt users to confirm they are well enough and would like to continue with the data donation cycle (see Figure 1a).

Maximizing Personal Agency During Data Donation. The loss of control over the disclosure of sexual experience can be detrimental to individuals with trauma [1, 13, 18]. Thus, our design prioritizes user agency through various affordances to control one’s donated data. These include: (1) users can “skip” any data donation prompt (see Figure 1b), and delete or edit past donations; (2) all donated data can be selectively covered by the user and will be automatically de-identified (see Figure 2); and (3) precautions are taken to prevent data donors from feeling labeled as SV perpetrators or victims. To the latter, we frame questions in a behaviorally specific manner without labeling anyone as a “victim” or perpetrator.”

4 DISCUSSION AND FUTURE WORK

Our research complements prior work into the stakeholder-inclusive design of data donation journeys [6, 12, 17] through the involvement of sexual violence experts who draw on their expertise to reflect on potential traumas of donating data about computer-mediated sexual violence.

Their emphasis on trauma-informed approach in cognitive walkthroughs of our data donation app draws parallels with HCI literature on trauma-informed computing and social media [10, 25] and demonstrates the value of extending this framework into human-centered AI. Our future work involves bridging insight from SV experts with feedback from anticipated data donors through usability testing.

ACKNOWLEDGMENTS

This material is based upon work partially supported by the U.S. National Science Foundation under Grant No. IIS-2211896.

REFERENCES

- [1] Christine E. Agaibi and John P. Wilson. 2005. Trauma, PTSD, and Resilience: A Review of the Literature. *Trauma, Violence, & Abuse* 6, 3 (Jul 2005), 195–216. <https://doi.org/10.1177/1524838005277438>
- [2] National Crime Agency. 2016. Emerging new threat in online dating: Initial trends in internet dating-initiated serious sexual assaults.
- [3] Shiza Ali, Afsaneh Razi, Seunghyun Kim, Ashwaq Alsoubai, Joshua Gracie, Munmun De Choudhury, Pamela J. Wisniewski, and Gianluca Stringhini. 2022. Understanding the Digital Lives of Youth: Analyzing Media Shared within Safe Versus Unsafe Private Conversations on Instagram. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 148, 14 pages. <https://doi.org/10.1145/3491102.3501969>
- [4] Ashwaq Alsoubai, Xavier V. Caddle, Ryan Doherty, Alexandra Taylor Koehler, Estefania Sanchez, Munmun De Choudhury, and Pamela J. Wisniewski. 2022. MOSafely, Is That Sus? A Youth-Centric Online Risk Assessment Dashboard. In *Companion Publication of the 2022 Conference on Computer Supported Cooperative Work and Social Computing* (Virtual Event, Taiwan) (CSCW'22 Companion). Association for Computing Machinery, New York, NY, USA, 197–200. <https://doi.org/10.1145/3500868.3559710>
- [5] Philip Anderson, Zheming Zuo, Longzhi Yang, and Yanpeng Qu. 2019. An Intelligent Online Grooming Detection System Using AI Technologies. In *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. 1–6. <https://doi.org/10.1109/FUZZ-IEEE.2019.8858973>
- [6] Karla Badillo-Urquiola, Zachary Shea, Zainab Agha, Irina Lediaeva, and Pamela Wisniewski. 2021. Conducting Risky Research with Teens: Co-Designing for the Ethical Treatment and Protection of Adolescents. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW3, Article 231 (jan 2021), 46 pages. <https://doi.org/10.1145/3432930>
- [7] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (Apr 2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [8] Rebecca Campbell, Rachael Goodman-Williams, and McKenzie Javorka. 2019. A Trauma-Informed Approach to Sexual Violence Research Ethics and Open Science. *Journal of Interpersonal Violence* 34, 23-24 (2019), 4765–4793. <https://doi.org/10.1177/0886260519871530> arXiv:<https://doi.org/10.1177/0886260519871530> PMID: 31514606.
- [9] Robey B. Champine, Jason M. Lang, Ashley M. Nelson, Rochelle F. Hanson, and Jacob K. Tebes. 2019. Systems Measures of a Trauma-Informed Approach: A Systematic Review. *American Journal of Community Psychology* 64, 3–4 (2019), 418–437. <https://doi.org/10.1002/ajcp.12388>
- [10] Janet X. Chen, Allison McDonald, Yixin Zou, Emily Tseng, Kevin A Roundy, Acar Tamersoy, Florian Schaub, Thomas Ristenpart, and Nicola Dell. 2022. Trauma-Informed Computing: Towards Safer Technology Experiences For All. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 544, 20 pages. <https://doi.org/10.1145/3491102.3517475>
- [11] Eric Filice, Kavishka D. Abeywickrama, Diana C. Parry, and Corey W. Johnson. 2022. Sexual violence and abuse in online dating: A scoping review. *Aggression and Violent Behavior* 67 (Nov 2022), 101781. <https://doi.org/10.1016/j.avb.2022.101781>
- [12] Alejandra Gomez Ortega, Jacky Bourgeois, and Gerd Kortuem. 2022. Reconstructing Intimate Contexts through Data Donation: A Case Study in Menstrual Tracking Technologies. In *Nordic Human-Computer Interaction Conference* (Aarhus, Denmark) (NordiCHI '22). Association for Computing Machinery, New York, NY, USA, Article 8, 12 pages. <https://doi.org/10.1145/3546155.3546646>
- [13] Eva Heim, Thanos Karatzias, and Andreas Maercker. 2022. Cultural concepts of distress and complex PTSD: Future directions for research and treatment. *Clinical Psychology Review* 93 (Apr 2022), 102143. <https://doi.org/10.1016/j.cpr.2022.102143>
- [14] Nicola Henry, Asher Flynn, and Anastasia Powell. 2020. Technology-Facilitated Domestic and Sexual Violence: A Review. *Violence Against Women* 26, 15–16 (Dec 2020), 1828–1854. <https://doi.org/10.1177/1077801219875821>
- [15] Larke N Huang, Rebecca Flatow, Tenly Biggs, Sara Afayee, Kelley Smith, Thomas Clark, and Mary Blake. 2014. SAMHSA's Concept of Trauma and Guidance for a Trauma-Informed Approach. (2014).
- [16] Younes Karimi, Anna Squicciarini, and Shomir Wilson. 2022. Automated Detection of Doxing on Twitter. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 276 (nov 2022), 24 pages. <https://doi.org/10.1145/3555167>
- [17] David Leimstädtner, Peter Sörries, and Claudia Müller-Birn. 2022. Unfolding Values through Systematic Guidance: Conducting a Value-Centered Participatory Workshop for a Patient-Oriented Data Donation. In *Proceedings of Mensch Und Computer 2022* (Darmstadt, Germany) (MuC '22). Association for Computing Machinery, New York, NY, USA, 477–482. <https://doi.org/10.1145/3543758.3547560>
- [18] Laura J. Long. 2022. Hope and PTSD. *Current Opinion in Psychology* 48 (Dec 2022), 101472. <https://doi.org/10.1016/j.copsyc.2022.101472>
- [19] Donna Maguire and James Taylor. 2019. A Systematic Review on Implementing Education and Training on Trauma-Informed Care to Nurses in Forensic Mental Health Settings. *Journal of Forensic Nursing* 15, 4 (Oct 2019), 242–249. <https://doi.org/10.1097/JFN.0000000000000262>
- [20] Afsaneh Razi, Ashwaq Alsoubai, Seunghyun Kim, Shiza Ali, Gianluca Stringhini, Munmun De Choudhury, and Pamela J. Wisniewski. 2023. Sliding into My DMs: Detecting Uncomfortable or Unsafe Sexual Risk Experiences within Instagram Direct Messages Grounded in the Perspective of Youth. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW1, Article 89 (apr 2023), 29 pages. <https://doi.org/10.1145/3579522>
- [21] Afsaneh Razi, Ashwaq Alsoubai, Seunghyun Kim, Nurun Naher, Shiza Ali, Gianluca Stringhini, Munmun De Choudhury, and Pamela J. Wisniewski. 2022. Instagram Data Donation: A Case Study on Collecting Ecologically Valid Social Media Data for the Purpose of Adolescent Online Risk Detection. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 39, 9 pages. <https://doi.org/10.1145/3491101.3503569>
- [22] Afsaneh Razi, Seunghyun Kim, Ashwaq Alsoubai, Gianluca Stringhini, Thamar Solorio, Munmun De Choudhury, and Pamela J. Wisniewski. 2021. A Human-Centered Systematic Literature Review of the Computational Approaches for Online Sexual Risk Detection. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 465 (oct 2021), 38 pages. <https://doi.org/10.1145/3479609>
- [23] Elizabeth Reeves. 2015. A Synthesis of the Literature on Trauma-Informed Care. *Issues in Mental Health Nursing* 36, 9 (Sep 2015), 698–709. <https://doi.org/10.3109/01612840.2015.1025319>
- [24] Mary Beth Rosson and John M Carroll. 2009. Scenario-based design. In *Human-computer interaction*. CRC Press, 161–180.
- [25] Carol F Scott, Gabriela Marcu, Riana Elyse Anderson, Mark W Newman, and Sarita Schoenebeck. 2023. Trauma-Informed Social Media: Towards Solutions for Reducing and Healing Online Harm. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 341, 20 pages. <https://doi.org/10.1145/3544548.3581512>
- [26] Rick Spencer. 2000. The streamlined cognitive walkthrough method, working around social constraints encountered in a software development company. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. ACM, The Hague The Netherlands, 353–359. <https://doi.org/10.1145/332040.332456>
- [27] Muhammad Uzair Tariq, Afsaneh Razi, Karla Badillo-Urquiola, and Pamela Wisniewski. 2019. A Review of the Gaps and Opportunities of Nudity and Skin Detection Algorithmic Research for the Purpose of Combating Adolescent Sexting Behaviors. In *Human-Computer Interaction. Design Practice in Contemporary Societies*, Masaaki Kurosu (Ed.). Springer International Publishing, Cham, 90–108.
- [28] Douglas Zytoko, Nicholas Furlo, Bailey Carlin, and Matthew Archer. 2021. Computer-Mediated Consent to Sex: The Context of Tinder. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 189 (apr 2021), 26 pages. <https://doi.org/10.1145/3449288>